

CGR-201 Command List

All commands must be terminated with a newline (0x0A)

Command	Command Syntax	Description
Identify	i	Returns an identification string and the firmware revision “*Syscomp CircuitGear MKII VX.X”
Scope Query state	?	Returns the current oscilloscope state machine state. 0 – Idle 1 – Pre-trigger sampling 2 – Waiting for trigger to reset 3 – Armed, waiting for trigger 4 – Triggered, post trigger sampling 5 – Reading data from RAM
Scope Start capture	c	Arms the scope for a capture. When the capture is complete, the scope will return a data stream with the following format: DA _H A _L B _H B _L A _H The ‘D’ character signifies the start of the data stream. A _H A _L are two bytes representing one 16-bit sample value from channel A. B _H B _L are two bytes representing one 16-bit sample value from channel B. The data stream is 4096 samples x 2 bytes/sample x 2 channels + 1 bytes long for a total of 16385 bytes. The data bytes are binary coded.
Scope Start strip chart sampling	C	Instructs the scope to start strip chart sampling
Scope Get strip chart sample	F	Request a strip chart sample. Returns ‘s’ if there is no sample ready. Returns 5 bytes: SA _H A _L B _H B _L The ‘S’ character signifies the start of the strip chart sample. A _H A _L are two bytes representing one 16-bit sample value from channel A. B _H B _L are two bytes representing one 16-bit sample value from channel B.
Scope Stop strip chart sampling	X	Instructs the scope to stop strip chart sampling

Scope Adjust channel Offset	o [A B] XXXX	Changes the DC offset for a scope input channel. A or B is the channel to adjust XXXX is an integer representing the offset from 0 to 4095 The offset integer is calculated as follows: $Offset = 2047 - \left(\frac{V_{offset}}{\frac{Step}{2}}\right)$ Where V_{offset} is the offset voltage. Step is the step size: 0.025 for the high input range, 0.002481 for the low input range.
Scope Select timbase	BX	Changes the timebase (sampling rate) X is an ascii character: 0 – 50ns/div to 5µs/div, 40 MHz sampling rate 1 – 10µs/div, 20 MHz sampling rate 2 – 20µs/div, 10MHz sampling rate 3 – 5 MHz sampling rate 4 – 50µs/div, 2.5MHz sampling rate 5 – 100µs/div, 1.25MHz sampling rate 6 – 200µs/div, 625 kHz sampling rate 7 – 500µs/div, 312.5 kHz sampling rate 8 – 1ms/div, 156.25kHz sampling rate 9 – 2ms/div, 78.125kHz sampling rate A – 5ms/div, 39.0625kHz sampling rate B – 10ms/div, 19.53125kHz sampling rate C – 20ms/div, 9.765625kHz sampling rate D – 4.8828125kHz sampling rate E – 50ms/div, 2.44140625kHz sampling rate F – 100ms/div, 1.2207403125kHz sampling rate G – 20ms/sample, strip chart mode H – 50ms/sample, strip chart mode I – 100ms/sample, strip chart mode J – 200ms/sample, strip chart mode K – 500ms/sample, strip chart mode L – 1s/sample, strip chart mode M – 2s/sample, strip chart mode

Scope DC Coupling	D [A B]	Selects DC coupling for channel A or B
Scope AC Coupling	A [A B]	Selects AC coupling for channel A or B
Scope Preamp range control	P [A B a b]	Selects preamp range: A – Channel A high range B – Channel B high range a – Channel A low range b – Channel B low range
Scope Enable auto-triggering	R	Enables auto-triggering
Scope Disable auto-triggering	r	Disables auto-triggering
Scope Set auto-trigger period	a YYYY	Sets the auto-trigger timer. YYYY is an integer. Timer period is YYYY x 32μs
Scope Manual trigger	M	Causes a manual trigger
Scope Set trigger level	T XXXX	Sets the trigger level. XXXX is an integer between 0 and 1024
Scope Rising trigger slope	+	Sets trigger slope to rising
Scope Falling trigger slope	-	Sets trigger slope to falling
Scope Channel A trigger source	S	Select channel A as trigger source
Scope Channel B trigger source	s	Select channel B as trigger source
Scope Select external triggering	Q	Sets external trigger input as trigger source
Scope Disable external triggering	q	Disables external trigger as trigger source

Scope Query trigger frequency	f	Returns a 3-byte number in the format: fXYZ The trigger frequency is given by: $\text{Frequency} = (X*65536 + Y*256 + Z)/2.56$
Digital I/O Set digital output	O XXX	Sets digital output to XXX XXX is a byte corresponding to the output bits
Digital I/O Read digital inputs	N	Returns the digital input state as "IX" Where 'I' is an ascii character X is a byte representing the current state of the digital inputs
Digital I/O Set digital input automatic read period	d YYYY	Sets the sampling period for automatically reading the digital inputs. YYYY is an integer The digital input sampling period is $YYYY \times 32\mu s$ The device will read the digital inputs each sampling period and return them to the host. Warning: fast sampling periods will swamp the host. Set the period to 0 to disable automatic sampling
Digital I/O Set PWM Frequency	G WWWXXXY YYZZZ	Sets the frequency of the PWM output. Same format/resolution as waveform output.
Digital I/O Set PWM Duty Cycle	Y XXX	Sets the PWM duty cycle to $XXX/255\%$. Where XXX is 0 to 255.
General Read USB Voltage	V	Reads the current USB voltage being supplied to the CircuitGear. Returns a two-byte 16-bit number in the format VXY where X is the high byte and Y is the low byte. The USB voltage is given by: $\text{Voltage} = 6.2 * (256 * X + Y) / 2047$
Waveform Generator Set frequency	WFWWWXX XYYYYZZZ	Set the DDS integer to WFWWWXXYYYYZZZ Each count is 0.04656613Hz
Waveform Generator Set Amplitude	WAYYYY	Set the waveform amplitude to $(YYYY/4095) \times 100\%$

Waveform Generator Set offset	WOYYYY	Adjust the output offset. YYYY is an integer between 0 and 4095 The offset integer YYYY is calculated as follows: $Offset = 4095 \left(\frac{(3 - V_{Offset})}{6} \right)$ Where <i>Offset</i> is the offset integer YYYY V_{Offset} is the output offset in volts
Waveform Generator Select waveform	WWX	Select the output waveform. X is an ascii character: 0 – Sine 1 – Square 2 – Triangle 3 – Sawtooth 4 – Custom 5 – Noise D - Disabled
Waveform Generator Load custom waveform buffer	WCXXXXYYY	Load a custom waveform sample into the device buffer. Each custom waveform is 2047 samples long consisting of 8-bit samples. XXXX is the sample index (0 to 2047) YYY is the sample value (0 to 256)
Waveform Generator Transfer waveform buffer to generator output	Wc	Applies the custom waveform buffer to the waveform generator output.
Waveform Generator Save custom waveform	WX	Saves the current custom waveform to non-volatile memory
Waveform Generator Read current waveform	WR	Reads the current waveform Returns RWS₀S₁...S₂₅₅ Where RW are ascii characters S₀ is the first sample (byte) S₂₅₅ is the last sample This command returns 258 bytes